

GENERAL SPECIFICATION G6.1

General

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G6.1 GENERAL

G6.1.1 General

- (a) This section of the specification details the general workmanship and materials requirements for the instrumentation, control and automation equipment required for the works.
- (b) Although they are specified in the mechanical equipment section of the Specification, final control elements such as valve actuators shall generally be included under the terms “instruments”, “ICA”, and “instrumentation” where these terms are used in relation to cabling, signals, I/O, and tag numbering.

G6.1.2 Degree of Protection

- (a) Where degrees of protection (IP ratings) are referred to, they refer to definitions given in the latest revision of BS EN 60529.

G6.1.3 Standards to be Complied with

- (a) In addition to standards listed elsewhere in the Specification, the latest edition of the following standards shall also be complied with:
 - (i) British Standards

Standard	Subject
BS 1553	Specification for graphical symbols for general engineering
BS EN 60073	Basic and safety principles for man-machine interface, marking and identification. Coding principles for indication devices and actuators.
BS EN 60529	Specification for degrees of protection provided by enclosures
BS 1042	Measurement of fluid flow in closed conduits
BS 1322	Specification for amino plastic moulding materials
BS 3406	Determination of Particle Size Distribution
BS 5424	Specification for Control gear for voltages up to and including 1000 Vac and 1200 Vdc
BS 5863	Analogue signals for process control systems
BS 6739	Code of practice for instrumentation in process control systems: installation design and practice.

(ii) American National Standards Institute / Instrumentation, Systems and Automation Society Standards

Standard	Subject
ANSI/ISA 5.1	Instrumentation symbols and identification
ANSI/ISA 5.3	Graphic Symbols for Distributed Control/Shared
ANSI/ISA 5.4	Instrument loop diagrams
ANSI/ISA 5.5	Graphic Symbols for Process Displays
ANSI/ISA 7.0.01	Quality Standard for Instrument Air
ANSI/ISA 26	Dynamic response testing of process control instrumentation
ANSI/ISA 50.1	Compatibility of analogue signals for electronic industrial process instruments
ANSI/ISA 50.02	Fieldbus Standard for use in Industrial Control Systems
ANSI/ISA 51.1	Process instrumentation terminology

(iii) Other Standards

Standard	Subject
IEC 60079-10	Classification of Hazardous Areas
IEC 60079-14	Electrical installations in hazardous areas
IEC 60079-17	Inspection and maintenance of electrical installations in hazardous areas
IEC 61131-3	Programmable Controllers - Part 3: Programming Languages
IEC 61158	Fieldbus for use in industrial control systems (technically equivalent to ANSI/ISA S50.02)
ISBN10 0118839063	United Kingdom Health & Safety Executive (HSE) Guidelines for Programmable Electronic Systems in safety related applications
Institution of Electrical Engineers	Guidelines for the Documentation of Computer Software for Real Time and Interactive Systems
IEC 60794 Part 1-1	Optical Fibre Cables Generic Specification – General
IEC 60794 Part 1-2	Optical Fibre Cables Generic Specification – Basic Optical Cable Test Procedures.
IEC62443	Industrial communication networks - Network and system security
IEC61508	Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems
IEC61511	Functional safety - Safety instrumented systems for the process industry sector

G6.1.4 Power Supply

- (a) Overall power supply to ICA systems shall be 230 VAC though individual components shall generally operate at 24 Vdc. Power supply to individual instruments shall generally be 24 Vdc as specified in General Specification G6.12 Instrumentation. Variation in supply voltage of $\pm 10\%$ and in frequency of $\pm 2\text{Hz}$ shall not measurably affect the operation or accuracy of instruments and all ICA equipment. Equipment shall function satisfactorily with transient voltage drops down to 85% of rated value.
- (b) Power supply to all ICA equipment (including control, monitoring, communication equipment, field instrumentation, analysers) and ICA panels shall be provided from a UPS backed source. UPS systems in turn shall have two independent sources of supply to provide redundancy.
- (c) Field instruments and solenoid valves shall, where models are available, operate on a 24 V dc power supply.
- (d) Where 24 V d.c. instruments are not available the Contractor shall submit a list of field instruments using other supply voltages to the S.O. for approval.
- (e) 3-pins utility socket type connections are not acceptable for connecting UPS units.
- (f) Refer to Electrical section in the Specification for complete detail of power supplies configuration.

G6.1.5 Process Connection

- (a) The process connections (i.e. thread type, size) for instruments shall be consistent with the other piping systems in use.

G6.1.6 Hazardous Areas

- (a) Hazardous Areas are areas where explosive gas/air/dust/fibre mixtures are likely to exist. These are defined in the Hazardous Area Classification and in Hazardous Area Zone Drawings.
- (b) ICA Equipment for installation in hazardous areas shall be selected and installed in accordance with IEC 60079.
- (c) All ICA equipment installed in hazardous areas shall use intrinsic safety Ex (i), as the method of protection. If Ex(i) is not available from the supply chain, then flameproof Ex(d) shall be the next preferred method of protection. If neither Ex(i) nor Ex(d) is available for a required item of ICA equipment, that needs to be installed in a hazardous area, a proposal to use a pressurised enclosure Ex(p) in accordance with IEC60079.2 may be submitted to the S.O. for consideration and approval.
- (d) All ICA equipment for hazardous areas shall be supplied with a certificate from the testing agency. One certificate for each model used shall be sufficient. Hard copies of certificates shall be arranged in a dedicated file. Copies shall be included in the Operation & Maintenance Manuals. The testing and certifying agency shall be ATEX/BASIEFA/UL/CSA/FM/CENELEC/ PTB.
- (e) Ex(d) apparatus shall display a warning on its outer cover stating “ WARNING - DO NOT OPEN UNTIL POWERED OFF”.

G6.1.7 Functional Safety and Safety Instrumented Functions

- (a) Instruments and devices detecting and reacting to conditions representing potential hazards to personnel shall be treated as safety instrumented functions (SIF's) and shall be categorised in accordance with IEC 60508 and designed in accordance with IEC61511.
- (b) Examples of instruments to which functional safety requirements may apply are: toxic or hazardous gas concentration monitors, flooding detection, emergency stop, biogas control valves etc.
- (c) Compliant solutions may involve combinations of the following, as appropriate to each SIF, and in accordance with the above listed standards:
 - (i) hard-wiring of the instrument signal to the warning device, protection circuit or switchgear;
 - (ii) the use of safety rated and certified electronic devices such as safety relays or safety PLCs.

G6.1.8 Software Factory Acceptance Test

- (a) The control system for each process, and any master overall plant controls shall be tested in a staging environment prior to loading the software into the live plant control system and site commissioning.
- (b) The staging environment test hardware shall include at least one each of the following hardware items:
 - (i) PLC CPU (hot standby pair if applicable);
 - (ii) Additional PLC CPU (hot standby pair if applicable) if test includes peer-to-peer communication;
 - (iii) Automation Network Sub-ring Switches (3);
 - (iv) Automation Network Backbone Switches (3);
 - (v) Monitoring Network Switches (3);
 - (vi) I/O Server (pair);
 - (vii) Remote I/O panel;
 - (viii) Motor Starter (DOL);
 - (ix) Motor Starter (VFD);
 - (x) SCADA Operator Workstation; and
 - (xi) Local HMI Panel (where applicable).

Additional quantities of hardware to match the software configuration may be made up by virtual device simulators.

- (c) The Contractor shall arrange for suitably experienced representatives from process system and equipment suppliers to be present during the testing to ensure the control system meets the all the process equipment requirements.
- (d) The Contractor shall notify the S.O. at least 4 weeks prior to commencement of the testing.

- (e) The software FAT shall include demonstrating and testing of every control function, alarm, interlock, etc. as well as testing and demonstrating all operator interfaces with the plant control.

G6.1.9 Components, Equipment and System Design

- (a) Environmental Considerations
 - (i) All equipment shall be suitable for the environmental conditions. Equipment mounted in enclosures shall be suitable for continuous operation at the maximum internal temperature possible in service, due account being taken of internally-generated heat and heat dissipated by other equipment. All components shall be rated adequately and circuits shall be designed so that change of component characteristics within the manufacturers' tolerances shall not affect the performance of plant. All equipment shall be designed to operate without artificial cooling.
 - (ii) Equipment provided with anti-condensation heaters shall be capable of operating without damage if the heaters are left on continuously. All equipment supplied shall be suitable for use in tropical climate.
- (b) Construction of Circuitry
 - (i) All equipment shall be designed for rapid fault-diagnosis and replacement of sub-assemblies and components;
 - (ii) Unless located in dedicated air-conditioned rooms, circuitry shall be tropicalized to operate continuously in temperatures ranging from 20 to 50 degrees centigrade with a Relative Humidity of up to 100 %;
 - (iii) Where specified elsewhere, printed circuit boards shall have conformal coating; and
 - (iv) All transformers shall be double-wound with an earthed screen between primary and secondary windings.
- (c) Components
 - (i) Solid-state devices shall be used in preference to moving-armature relays;
 - (ii) Relays shall be of the plug-in type with an indicating LED and shall be hermetically sealed; and
 - (iii) Components shall additionally comply with General Specification G1.3 Workmanship Requirements.
- (d) Effect of Failures
 - (i) The control system shall be designed so that failures of components or power supplies shall initiate alarms or protective actions and not lead to any potentially hazardous condition or mask any actual alarm or unhealthy state.
 - (ii) Plant protection and control systems shall be designed so that their outputs are de-energised or neutralised whenever a failure occurs of any component or circuit or power supply associated with that protection or control circuit.
 - (iii) No single equipment fault shall prevent the correct operation of any protection or shut-down circuit whenever such operation is necessitated by a plant fault condition or control action.

- (iv) Under emergency, failure or shut-down circumstances, each regulating device shall move to the appropriate safe condition or stay-put as required.
- (e) Immunity to Interference
 - (i) All devices shall be immune to normal industrial interference, mains borne interference and to radio transmitting devices up to 2000 MHz. Transmitters of 5 watts RF power at up to 500 MHz and 1 watt at up to 2000 MHz can be expected within 1 metre proximity.
- (f) Proven Equipment
 - (i) All equipment, shall be of proven performance in water treatment and used water treatment environments.
 - (ii) All equipment, including software, shall be locally supported with fully capable locally based engineering and technical expertise. Local resources shall be experienced in the detailed technical and engineering aspects of the equipment and software.
 - (iii) In the case of SCADA software, PLC software, and plant historian software, it shall be a requirement that there is at least one fully operational installation, at a comparable scale, in Singapore of the software being proposed by the Contractor.
- (g) Visibility and Readability of Displays
 - (i) All displays shall be clearly visible and readable in the lighting conditions that would normally be expected in each location. In this context, “displays” shall include but not be limited to panel lamps, LCD displays of any size, LED displays of any size, CRT displays of any size, and fluorescent displays of any size.
 - (ii) The Contractor shall ensure that all displays are provided such that they are clearly visible and readable in all expected lighting conditions. It shall not be acceptable for the operators to need to shade the display with their body to view a display.

G6.1.10 Telephone Connections

- (a) Refer to General Specification G7 Building Services.

G6.1.11 Supply Chain Verification

- (a) All equipment supplied shall be sourced from local authorised distributors within Singapore, carrying the full backing and support of the manufacturer.
- (b) If requested by the S.O. the Contractor shall provide documentary evidence of the supply chain.

G6.1.12 Control System Functional Requirements

- (a) The system shall exhibit the highest degree of availability, user-friendliness, reliability and maintainability. The design shall include, but not be limited to, the following features:
 - (i) The system shall be designed to recover fully from hardware, software and power supply failures in a bumpless manner;

- (ii) The system shall be protected against cascading component failures;
- (iii) Duty standby equipment and instruments shall be physically connected to separate field networks and/or separate I/O units or modules;
- (iv) The system shall reject illegal instructions, inconsistent data, inadvertent actions and unauthorized controls; and
- (v) No inadvertent indication or incorrect control shall occur under any or all of the following conditions:
 - 1) Single component failures;
 - 2) Failure or replacement of any fuse;
 - 3) Replacement of a plug-in processor, power supply or communications module;
 - 4) Transient signal noise on signal and data circuits;
 - 5) Operation during extreme service conditions and over noisy communications lines;
 - 6) Initial switch-on or restoration after component failure, power supply failure, testing or maintenance;
 - 7) Software development and debugging; and
 - 8) Modification and expansion of facilities.
- (b) The control system shall be designed to support on-line editing of program without the need to shut down the controller for program compilation.

END OF SECTION